

Industry-Oriented NumPy Problem Statements with Integrated Mini Project Extensions

Note: Mini project extensions part is optional. Advanced learners are encouraged to carryout mini project.

1. Manufacturing Analytics

Title

Factory Production Analysis and Performance Monitoring System Using NumPy

Problem Statement

A manufacturing company operates **4 production lines across 5 shifts** every day. The management wants to analyze production efficiency and identify underperforming production lines.

Create a **4 × 5 NumPy array** representing production units manufactured by each production line during different shifts.

Perform the following operations:

1. Calculate the mean production units across all shifts and production lines.
2. Calculate the standard deviation to measure production variability.
3. Slice the first 3 production lines and the last 2 shifts.
4. Reshape the array into a **5 × 4 matrix**.
5. Create another **5 × 4 machine efficiency matrix** and perform element-wise multiplication.

Mini Project Extension

Develop a **Factory Performance Analytics Dashboard** that:

- Monitors production efficiency in real time.
- Compares production lines.
- Identifies low-performing shifts.

- Generates production reports.
- Predicts maintenance schedules.
- Visualizes production trends using charts and heatmaps.

Learning Outcome

Students learn how matrix operations can be applied to industrial production monitoring and decision-making.

2. Transportation Analytics

Title

Smart Traffic Flow Analysis and Congestion Monitoring Using NumPy

Problem Statement

A municipal corporation monitors traffic flow at **5 intersections during 6 time intervals** daily.

Create a **5 × 6 NumPy traffic matrix** representing vehicle counts.

Perform the following:

1. Calculate the average traffic volume.
2. Determine the standard deviation of traffic counts.
3. Extract the first 2 intersections and the last 3 time intervals.
4. Reshape the matrix into a **10 × 3 matrix**.
5. Perform element-wise multiplication with a congestion-weight matrix.

Mini Project Extension

Develop a **Traffic Analytics Dashboard** that:

- Identifies traffic congestion hotspots.
- Detects peak-hour traffic patterns.
- Recommends route diversions.
- Supports signal optimization.
- Generates traffic management reports.

Learning Outcome

Students apply NumPy-based matrix operations to traffic planning and smart city initiatives.

3. Renewable Energy Analytics

Title

Solar Farm Energy Production Analysis and Monitoring Using NumPy

Problem Statement

A renewable energy company records power generation from **6 solar farms across 4 time periods**.

Create a **6 × 4 NumPy matrix** representing energy generated.

Perform:

1. Calculate the average energy generated.
2. Determine the standard deviation of energy production.
3. Extract the first 4 farms and last 2 periods.
4. Reshape the matrix into a **3 × 8 matrix**.
5. Multiply by a solar efficiency matrix of the same shape.

Mini Project Extension

Develop a **Solar Energy Analytics Dashboard** that:

- Tracks energy generation trends.
- Compares farm efficiency.
- Identifies underperforming farms.
- Generates sustainability reports.
- Supports renewable energy planning.

Learning Outcome

Students learn how numerical analytics supports sustainable energy management.

4. Sports Analytics

Title

Athlete Performance Evaluation and Training Analytics Using NumPy

Problem Statement

A sports academy tracks performance scores of **8 athletes across 5 training sessions**.

Create an **8 × 5 matrix** representing athlete scores.

Perform:

1. Calculate average performance score.
2. Calculate standard deviation.
3. Extract the first 3 athletes and last 2 sessions.
4. Reshape into a **10 × 4 matrix**.
5. Multiply by a training intensity matrix.

Mini Project Extension

Develop an **Athlete Performance Dashboard** that:

- Tracks athlete progress.
- Compares training performance.
- Identifies top-performing athletes.
- Generates coaching reports.
- Recommends personalized training plans.

Learning Outcome

Students learn performance analytics and sports data evaluation techniques.

5. Aviation Analytics

Title

Flight Delay Analysis and Airline Operations Dashboard Using NumPy

Problem Statement

An airline records delay information for **7 routes across 4 operational periods**.

Create a **7 × 4 NumPy matrix** representing delay times.

Perform:

1. Compute average delay.
2. Calculate standard deviation.
3. Slice the first 3 routes and last 2 periods.
4. Reshape into a **4 × 7 matrix**.
5. Multiply by operational impact coefficients.

Mini Project Extension

Develop a **Flight Operations Analytics Dashboard** that:

- Monitors route performance.
- Detects recurring delays.
- Supports scheduling optimization.
- Generates operational reports.
- Improves airline efficiency.

Learning Outcome

Students understand how analytics supports aviation operations.

6. Movie Streaming Analytics

Title

Viewer Engagement and Content Analytics Using NumPy

Problem Statement

A streaming platform records viewer engagement for **5 genres across 8 weeks**.

Create a **5 × 8 matrix** of engagement scores.

Perform:

1. Calculate mean engagement.
2. Calculate standard deviation.
3. Extract the first 2 genres and last 3 weeks.
4. Reshape into a **10 × 4 matrix**.
5. Multiply by a popularity-score matrix.

Mini Project Extension

Develop a **Content Recommendation Dashboard** that:

- Identifies trending genres.
- Tracks viewer preferences.
- Supports content recommendation decisions.
- Generates weekly engagement reports.
- Predicts content demand.

Learning Outcome

Students learn data-driven content analytics and recommendation systems.

7. Telecommunication Analytics

Title

Network Utilization and Optimization Analytics Using NumPy

Problem Statement

A telecom provider records data usage across **6 regions and 6 time slots**.

Create a **6 × 6 network usage matrix**.

Perform:

1. Calculate average usage.
2. Determine variability using standard deviation.
3. Extract the first 3 regions and last 2 slots.
4. Reshape into a **9 × 4 matrix**.
5. Multiply by bandwidth-capacity factors.

Mini Project Extension

Develop a **Network Optimization Dashboard** that:

- Monitors bandwidth utilization.
- Detects overloaded regions.
- Recommends capacity upgrades.
- Generates network reports.

- Supports infrastructure planning.

Learning Outcome

Students understand telecom performance analysis.

8. Education Technology (EdTech) Analytics

Title

Learning Participation and Course Analytics Using NumPy

Problem Statement

An online learning platform records participation across **10 courses during 3 months**.

Create a **10 × 3 participation matrix**.

Perform:

1. Calculate average participation.
2. Calculate standard deviation.
3. Extract the first 5 courses and last 2 months.
4. Reshape into a **5 × 6 matrix**.
5. Multiply by course difficulty scores.

Mini Project Extension

Develop a **Learning Analytics Dashboard** that:

- Tracks learner engagement.
- Identifies popular courses.
- Detects low-participation courses.
- Supports curriculum improvement.
- Generates educational analytics reports.

Learning Outcome

Students gain experience in educational data analytics.

9. Logistics and Supply Chain Analytics

Title

Warehouse Shipment and Logistics Analytics Using NumPy

Problem Statement

A logistics company tracks shipments from **4 warehouses across 7 weeks**.

Create a **4 × 7 shipment matrix**.

Perform:

1. Compute average shipments.
2. Compute standard deviation.
3. Slice the first 2 warehouses and last 3 weeks.
4. Reshape into a **7 × 4 matrix**.
5. Multiply by transportation cost matrices.

Mini Project Extension

Develop a **Supply Chain Analytics Dashboard** that:

- Tracks inventory movement.
- Optimizes transportation costs.
- Identifies logistics bottlenecks.
- Generates warehouse performance reports.
- Supports supply chain planning.

Learning Outcome

Students understand numerical analytics in logistics management.

10. Space Research Analytics

Title

Satellite Sensor Monitoring and Mission Analytics Using NumPy

Problem Statement

A space agency records sensor readings from **9 satellites across 4 observation periods**.

Create a **9 × 4 sensor matrix**.

Perform:

1. Calculate mean sensor reading.
2. Calculate standard deviation.
3. Extract the first 4 satellites and last 2 observations.
4. Reshape into a **6 × 6 matrix**.
5. Multiply by calibration coefficient matrices.

Mini Project Extension

Develop a **Satellite Health Monitoring Dashboard** that:

- Tracks satellite sensor performance.
- Detects anomalies.
- Supports mission planning.
- Generates space operations reports.
- Monitors spacecraft health metrics.

Learning Outcome

Students learn aerospace analytics and scientific computing applications.
